

VJ X-ray A VJ Technologies Company	SPEC NUMBER: P562 SHT 1 OF 7	REV: 1
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FILES ASSOCIATED WITH THIS SPECIFICATION

FILENAME	CONTENTS
P562-IXS-FIRMWARE-P562 R1.docx	This Document

CHANGE HISTORY

ORIGINATOR	DATE	DESCRIPTION OF CHANGE
Jesse Stoisch	2020.7.15	Initial Release - Reference Model P323-R1.07 - Final Fix 323 Updated
Jesse Stoisch	2023.1.09	Rev 1 - Fixed WDTE command timing to 5 sec to reset Watchdog

TITLE: FIRMWARE-P562 Specification	
GENERATED BY: Jesse Stoisch	DOC OWNER: X-ray Eng. & Sales
REVIEWED BY: Joseph Zhou	REVIEW DATE: 1/9/23
APPROVED BY: NY Eng. Team	APPROVAL DATE:

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DIGITAL COMMUNICATION PROTOCOL

1. RS232 DIGITAL INTERFACE

Overview

RS232 based Digital Interface between user and the X-Ray Controller. It provided the ability to program the output voltage and current, monitor the output voltage and current, and monitor the fault conditions.

1.1 RS232 Communication Parameters

The RS232 interface has the following attributes:

9600 baud

No Parity

8 Data Bits 1 Stop

Bit

XON / XOFF handshaking (also called software handshaking) is not supported. Hardware handshaking is not supported. None of the RS232C signals DTR, DSR, CTS, RTS, RI, or DCD is supported.

(Connect an RS232 (9 pin) M/F straight cable between Computer RS232 port to J3 on the XRay control.)

1.2 Command Structure

There are two categories of commands from the host computer to the tank. 1)

Commands that have an argument.

2) Commands that do not have an argument.

The syntax of commands that have an argument is:

<STX>CMDARG<CR>

The syntax of commands that do not have an argument is:

<STX>CMD<CR>

The specification of the above symbols is as follows:

<STX> Start of message. The hex value 0x02 is used.

CMD One of the commands defined below.

ARG An argument string that is defined below for each command. <CR> The carriage return character 0x0D.

1.3 Report Structure

Reports sent from the power supply to the host computer shall be structured as an ASCII string as shown below: <STX>RPT<CR>

<STX> Start of message. The Hex value 0x02 used to indicate the start of a message.

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RPT Report. The reports are defined below.
 <CR> The carriage return character 0x0D.
 <SP> The Space Character 0x20.

1.4 Command Arguments

Command arguments (when used) are always 1 to 8 characters representing a number.

1.5 Command and Report Definitions:

Command and Report Definitions				
Name	Command	Argument	Response	Comments
Voltage Program	VP	XXXX	<STX>VPXXXX<CR>	XXXX = 000.0 – Max KV
Current Program	CP	XXXXX	<STX>CPXXXXX<CR>	XXXXX = 00000 – Max uA
Voltage/Current/Temperature/ Filament Monitor	MON	N/A	<STX>VVVV<SP>CCCC <SP>TTTT<SP> FFFF<CR>	VVVV = 000.0–Max KV CCCC = 00000 – Max uA TTTT = 000.0 – 070.0 DegC FFFF = 0000 – 4095
Fault Clear	CLR	N/A	<STX>CLR<CR>	
Report Fault	FLT	N/A	<STX>X<SP>X<SP>X<SP> >X<SP>X<SP>X<SP>X<S P>X<SP>X<SP>X<CR>	X X X X X X X X X = See Below
HV Status	STAT	N/A	<STX>X<CR>	X = 1 is X-Ray On X = 0 is X-Ray Off
X-Ray Enable	ENBL	X	<STX>ENBLX<CR>	X = 1 Enable X-Ray X = 0 Disable X-Ray
Enable / Disable X-Ray Minimization	MIN	X	<STX>MINX<CR>	X = 1 Enable Minimization X = 0 Disable Minimization
Watch Dog Timer	WDTE	N/A	<STX>OK<CR>	
Comm Port Echo	FREV	N/A	<STX>XNNN<CR>	XNNN = 2000
Watch Dog Enable/Disable	WDOG	X	<STX>WDOGX<CR>	X = 1 Enable Watch Dog X = 0 Disable Watch Dog
Watch Dog Status	WSTAT	N/A	<STX>X<CR>	X = 1 Watch Dog On X = 0 Watch Dog Off
X-Ray Minimization Status	MSTAT	N/A	<STX>X<CR>	X = 1 Minimization is On X = 0 Minimization is Off
Serial Number Info	SNUM	N/A	<STX>XXXXXXXXXXXXX< CR>	XXXXXXXXXXXXX = Serial Number
Model Number Info	MNUM	N/A	<STX>MMMMMMMMMMMM MMMM<CR>	MMMMMMMMMMMMMM M = Model Number
Pre-warn Status	PSTAT	N/A	<STX>X<CR>	X = 1 Pre-warn Stat On X = 0 Pre-warn Stat Off

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Pre-warn Time Set	PTM	XX	<STX>PTMXX<CR>	XX = 0-60 Seconds
Pre-warn Time Selected Status	PTST	N/A	<STX>XX<CR>	XX = 0-60 Seconds
Cumulative X-Ray On Time	XTM	N/A	<STX>XXXXXX<SP>YY<CR> >	XXXXXX = hours YY = minutes
Cumulative Power On Time	POT	N/A	<STX>XXXXXX<SP>YY<CR> >	XXXXXX = hours YY = minutes
Reset All Timers	RST	XXXXXX	<STX>RSTXXXXXX<CR>	XXXXXX = X-Ray Hours to set to, all others are set to zero
Get Oil Temperature Flash Log	TLOG	N/A	<STX>XXXXXX<SP>YY <SP>ZZZZ<CR> 32 Times	XXXXXX = Power On-Time Hours YY = Power On-Time Minutes ZZZZ = Oil Temperature at that Power On-Time (000.0-070.0 Deg C)

1.6 X-Ray Minimization Command

This command enables/disables the X-Ray Minimization feature. X-Ray Minimization is active only when X-Ray is enabled. Upon X-Ray Minimization, the X-Ray Output is minimized by setting mA and Filament DAC's to 0V by the X-Ray Controller. Disabling X-Ray Minimization will return the X-Ray Output to its commanded output.

1.7 X-Ray Minimization Status

This command reflects the Active status of X-Ray Minimization. X-Ray Disable will automatically disable the X-Ray Minimization.

1.8 WDOG Command

This command enables/disables the Watch Dog feature. Upon Power up X-Ray Generator has Watch Dog feature Enabled. Once WDOG Disable command is sent to X-Ray Controller then it turns off Watch Dog feature until next Power Cycle.

1.9 WDTE Command

Once the Microcontroller doesn't receive any command within 5s after it sends response to previous command then it turns off X-Ray Enable and turns off KV program and mA Program. So WDTE command should be sent once at least 5s if no command is sent to the Microcontroller.

NOTE: If Watch Dog feature is disabled then this command has no effect on X-Ray Enable Status.

1.10 XTM Command

This command gets the cumulative X-Ray on time in hours and minutes. It indicates the cumulative X-Ray on time for the unit as a whole. It gets updated every minute when the X-ray

is turned on, and when the X-ray is turned off after being on. This cumulative X-Ray on time is saved in a circular buffer of up to 32 locations in non-volatile flash memory.

1.11 POT Command

This command gets the cumulative on time for how long the unit has been powered up, in hours and minutes. It indicates the cumulative power on time for the unit as a whole. It gets updated every time the X-Ray On Time gets updated, when the X-Ray is on, and every hour when the X-Ray is off. This cumulative power on time is saved in a circular buffer of up to 32 locations in non-volatile flash memory

1.12 Get Oil Temperature Flash Log Command

Each time an update happens for either on time the oil temperature is also saved at that time. Each entry is saved in a circular buffer of up to 32 locations in non-volatile memory. For each entry, the power on time serves as the timestamp for that temperature. This command gets the log of the temperature by returning the contents of that buffer's 32 locations.

1.13 Reset All Timers

This command resets all the contents of the circular buffer to zero, or sets just the X-Ray On Time entries to a specific number of hours and all the rest to zero. You reset the time by sending all zeros for its hours argument or set to a larger number than zero for setting X-ray On Time to specific number of hours.

1.14 Serial Command Handling

Command Response Time:

All commands from the host will be processed and replied anywhere from less than 100ms plus output ramp time.

1.15 Comm Port Echo:

This command is used to automatically configure Communication Port by user GUI. GUI can send this command to each available Comm Port in computer and when GUI receives response from X-Ray Controller, GUI can establish a link with Comm Port.

1.16 Command Buffering:

The X-Ray tank does not buffer commands. The tank can process only one command at a time. Before sending another command, wait for the response of the previous command.

1.17 Report Fault Definitions

Fault Bit(X9 X8 X7 X6 X5 X4 X3 X2 X1 X0)	Description
X0	X0 = 1 Regulation X0 = 0 No Regulation Fault
X1	X1 = 1 Interlock Open X1 = 0 No Interlock Open

X2	X2 = 1 Cathode Over KV Fault X2 = 0 No Fault
X3	X3 = 1 Anode Over KV Fault X3 = 0 No Fault
X4	X4 = 1 Over Temperature Fault X4 = 0 No Fault
X5	X5 = 1 Arc Fault X5 = 0 No Fault
X6	X6 = 1 Over Current Fault X6 = 0 No Fault
X7	X7 = 1 Power Limit Fault X7 = 0 No Fault
X8	X8 = 1 Over Voltage Fault X8 = 0 No Fault
X9	X9 = 1 Duty Cycle mode ON X9 = 0 Duty Cycle mode OFF

Fault Cause and Effect:

If a shutdown occurred from a fault, the fault must be cleared before X-Ray On can be enabled.

When a fault occurs, RS232 command "CLR" must be used to clear the fault. Once cleared, a 'High' on the enable signal to the micro controller will turn X-Rays On.

Fault conditions will not cause the tank to forget any stored information, such as the tank voltage and current. For example, if the voltage is set to 150 kV and a fault then occurs, the voltage will still be set to 150 kV after the fault is cleared.

Fault Definition:

- a. **Regulation Fault:** When kV or mA is no longer in regulation. This will disable the high voltage output.
- b. **Interlock Fault:** When J2-2 does not receive an Interlock input voltage (15VDC or 24VDC as per the model Specification). This will disable the high voltage output.
- c. **Cathode Over KV Fault:** When Cathode output voltage exceeds the maximum rated output by 5% to 10%. This will disable the high voltage output (Max Duration of over voltage before shutdown is 100ms).
- d. **Anode Over KV Fault:** When Anode output voltage exceeds the maximum rated output by 5% to 10%. This will disable the high voltage output (Max Duration of over voltage before shutdown is 100ms)

- e. **Over temperature Fault:** when Oil temperature reaches 60 ± 3 °C. This will disable the high voltage output.
- f. **Arc Fault:** When four (4) Arcs occurred within 10Sec. This will disable the high voltage output.
- g. **Over-current Fault:** When output current exceeds the maximum rated output by 5% to 10%. This will disable the high voltage output (Max Duration of over current before shutdown is 100ms).
- h. **Power Limit Fault:** When the kV/mA feedback values exceed 5% to 10% of the maximum rated power, this will disable the high voltage output.
- i. **Duty Cycle protection: (This item only applies to units that operate under pulse mode)**
 - The unit does not accept kV and mA program values above the maximum peak kV, mA and power ratings.
 - The unit does not turn On before completing the required Off time as per below Duty cycle definitions.
 - When the “Duty cycle mode” indicator is On, the unit does not accept new command. The unit does not accept new KV and mA program values during On time.
 - Duty cycle is defined such:
 Max duration = Max On time + Min Off time = Tmax (Tmax is defined as a fix value)
 Max continuous power = Pcont. (Pcont is defined as a fix value)
 Max peak power = Ppk-max. (Ppk-max is defined as a fix value)

Max requested peak power (Preq) = kV x mA (kV/mA are the program values)

$$T(\text{on}) = (P_{\text{cont}} \times T_{\text{max}}) / P_{\text{req}}$$

$$T(\text{off}) = T_{\text{max}} - T(\text{on})$$

Example:

$$\begin{array}{ll} T_{\text{max}} = 30\text{Sec} & P_{\text{cont}} = 200\text{W} \\ P_{\text{pk-max}} = 500\text{W} & P_{\text{req}} = 400\text{W} \end{array}$$

Then:

$$\begin{array}{l} \text{For } P_{\text{req}} = 400\text{W} \quad (400\text{W} \leq 500\text{W}) \\ T(\text{on}) = (200\text{W} \times 30\text{S}) / 400\text{W} = 15\text{Sec} \\ T(\text{off}) = 30\text{Sec} - 15\text{Sec} = 15\text{Sec} \end{array}$$

$$\begin{array}{l} \text{For } P_{\text{req}} = 500\text{W} \\ T(\text{on}) = (200\text{W} \times 30\text{S}) / 500\text{W} = 12\text{Sec} \\ T(\text{off}) = 30\text{Sec} - 12\text{Sec} = 18\text{Sec} \end{array}$$

2. ETHRNET INTERFACE

All commands and replies are the same on serial or Ethernet. User can connect to TCP with the following address:

IP Address: 192.168.10.120

Port: 10001

Connection Type: raw

End of Document